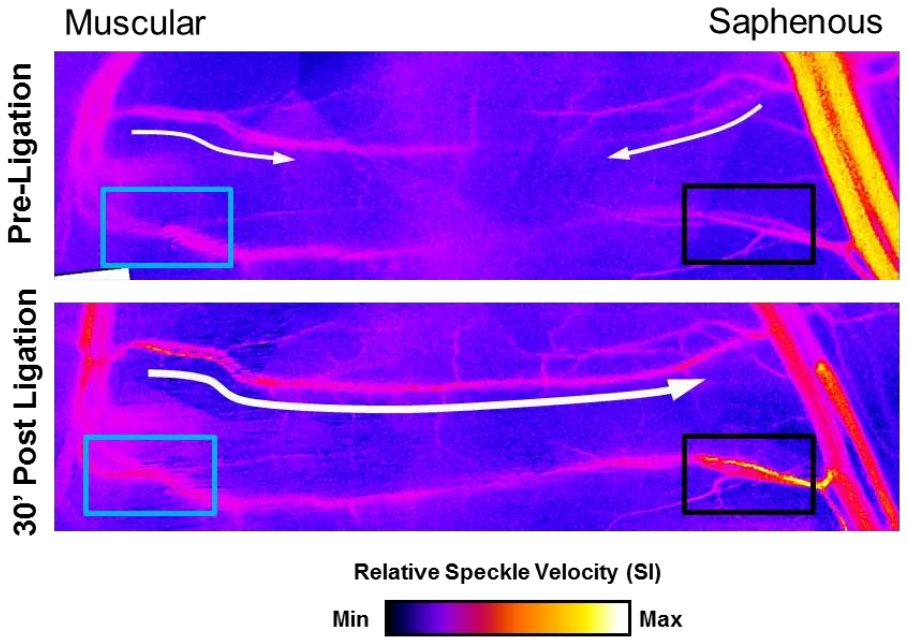
BME 2315: Computational Biomedical Engineering
Gaussian Elimination



Announcements

- **Problem Set 5 due next Tuesday**
- **drop deadline is 2/27**

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Adapted from Jeff Saucerman

Brief review of linear algebra basics

- I really recommend you watch the 3blue1brown series to get an intuitive understanding of matrices as linear transformations
- Vectors: “arrows pointing in space” or ordered list of numbers → two ways to think about vectors
- Matrices: linear transformations of vectors
- Systems of equations can be represented as a matrix

https://www.youtube.com/watch?v=fNk_zzaMoSs&list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab

Chapters 1-6 and 12 (optional)

Matrices and Vectors

Matrix

$$a = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}$$

Vectors

$$a = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \end{bmatrix}$$

$$a = \begin{bmatrix} a_{1,1} \\ a_{2,1} \\ a_{3,1} \end{bmatrix}$$

Determinant of a Matrix

Determinant summarizes a square matrix: $\det(A) = |A|$

For 2x2: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $|\mathrm{A}| = \mathrm{ad} - \mathrm{bc}$

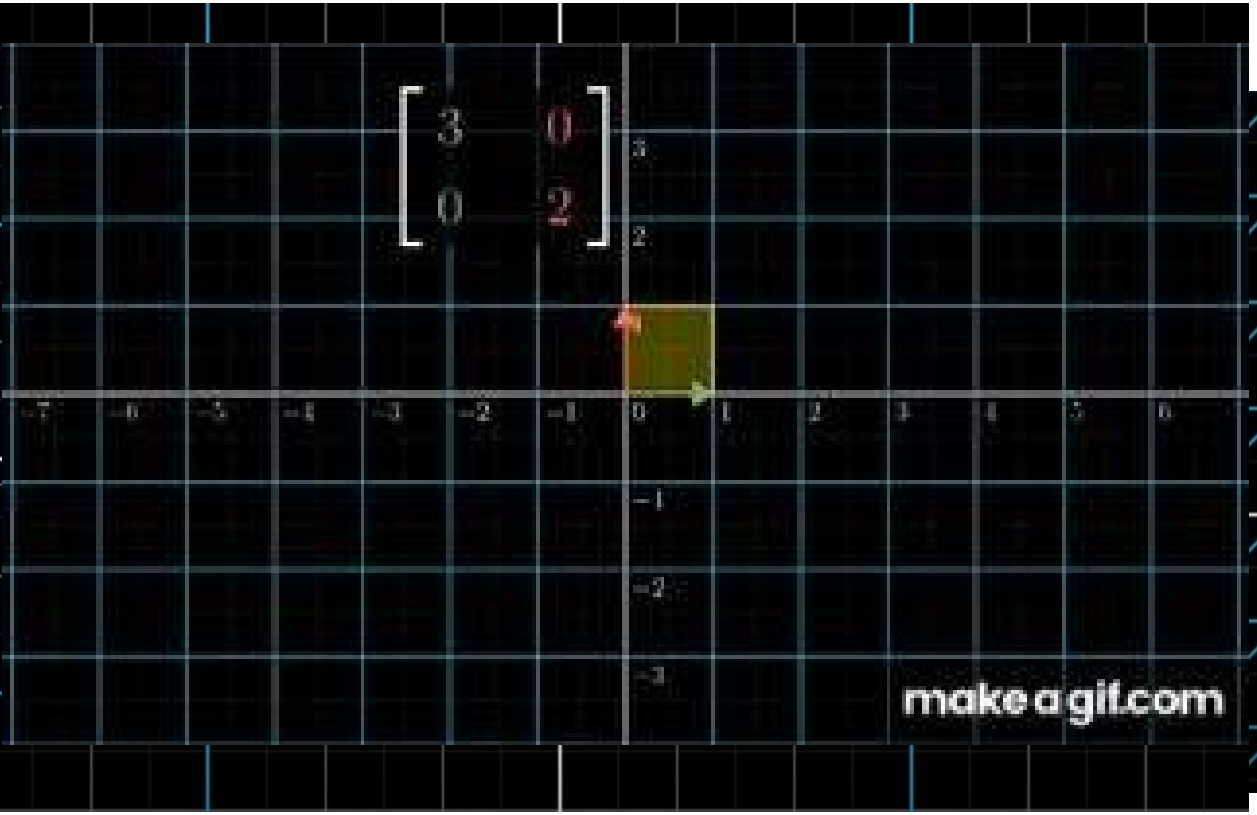
For 3x3: $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$,

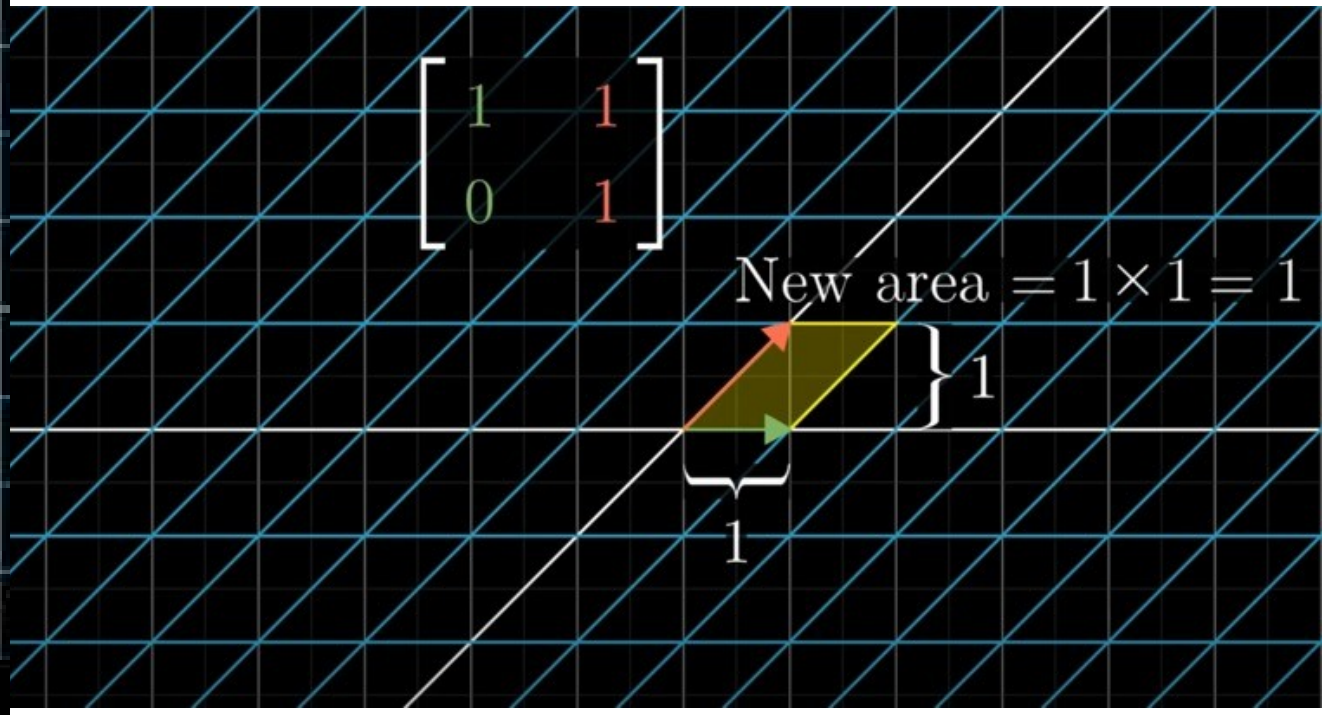
$$|\mathrm{A}| = (\mathrm{aei} + \mathrm{bfg} + \mathrm{cdh}) - (\mathrm{ceg} + \mathrm{bdi} + \mathrm{afh})$$

If $|\mathrm{A}| = 0$, the matrix is *singular*

A = $\begin{bmatrix} 3 & 1 & -7 \\ -2 & 0 & 4 \\ 9 & -1 & 8 \end{bmatrix}$. What is the determinant of A?

Determinant, intuitively





Matrix Multiplication

2 matrices:

$$C = A \cdot B$$

h x m h x n n x m

$$C = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \\ a_{3,1} & a_{3,2} \end{bmatrix} \begin{bmatrix} b_{1,1} & b_{1,2} \\ b_{2,1} & b_{2,2} \end{bmatrix} = \begin{bmatrix} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \\ c_{3,1} & c_{3,2} \end{bmatrix}$$

c(row i, column j) = sum [a(row i) .* b(column j)]

Inner dimensions must agree!

$$c_{i,j} = \sum_{k=1}^{n} a_{i,k} \cdot b_{k,j}$$

https://www.youtube.com/watch?v=XkY2DOUCWMU&list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab&index=4

Vector and Matrix Norms

Magnitude of a vector

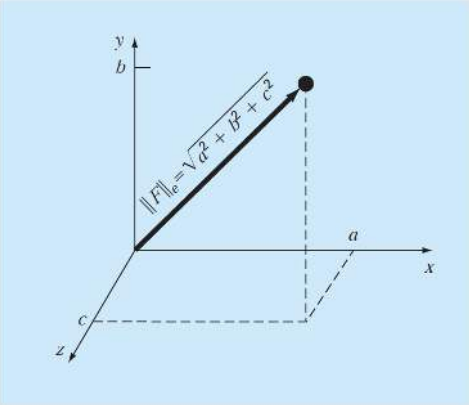
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\text{Length} = \sqrt{{x_1}^2 + {x_2}^2 + {x_3}^2}$$

$$= \sqrt{[X]^T[X]} = \|X\| \quad \text{“Norm”}$$

Matrix norm

$$\|A\| = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} {a_{ij}}^2}$$



Vector norm as Euclidean distance

$$\|\mathbf{x} - \mathbf{y}\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_N - y_N)^2}$$

$$\cos\theta = \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}$$

*there are many other types of norms. These are technically Euclidean and Frobenius norms, respectively. Here we’ll just say ‘norm’.

Representing systems of equations as a matrix

$$
\begin{aligned}
2 \cdot x_1 + 1 \cdot x_2 - 3 \cdot x_3 &= 8 \\
-3 \cdot x_1 \qquad\quad + 2 \cdot x_3 &= -2 \\
8 \cdot x_1 - 1 \cdot x_2 + 4 \cdot x_3 &= 12
\end{aligned}
$$

Solving Systems of Linear Equations

Conservation equations for complicated systems

$$[A][x] = [b]$$

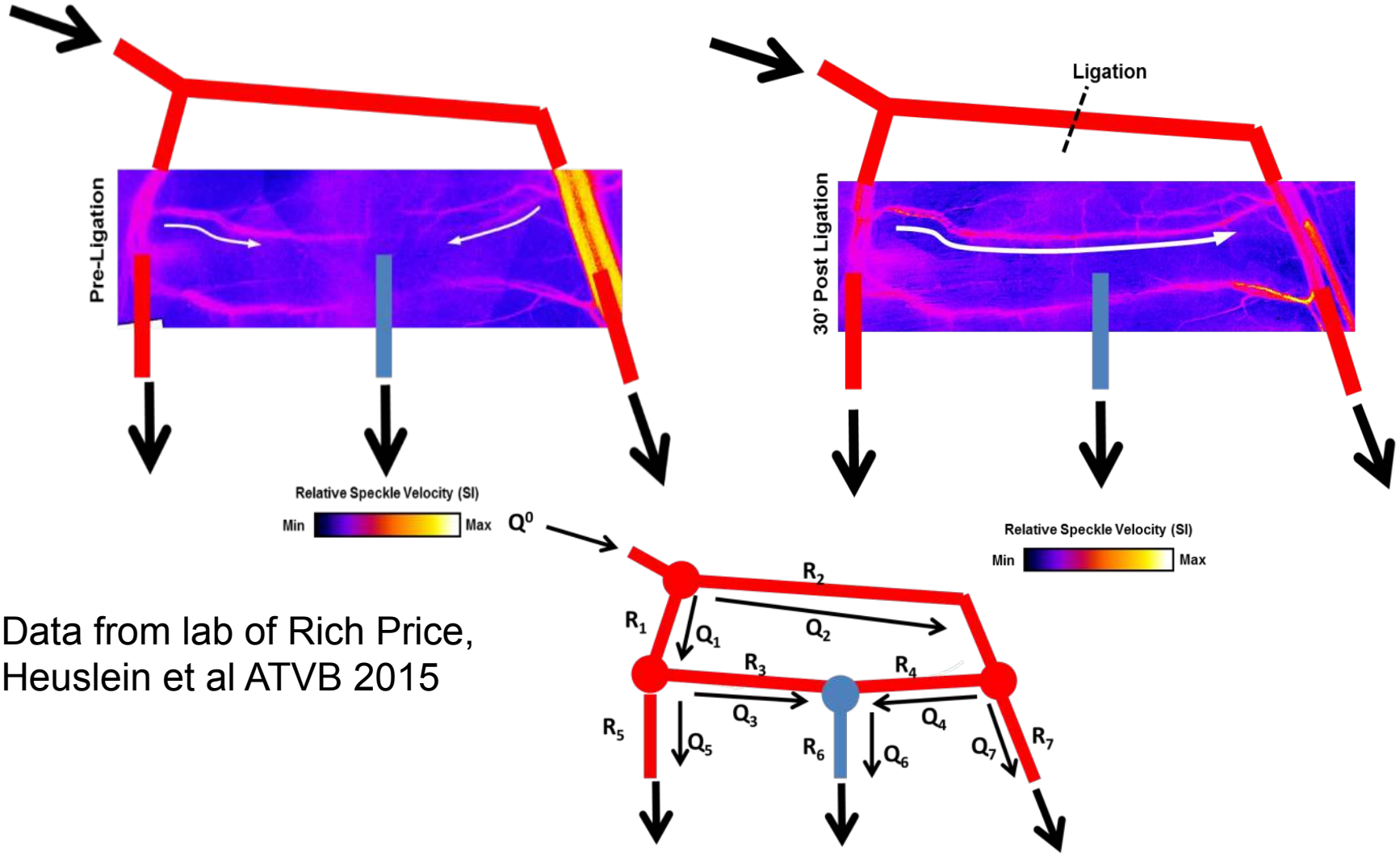
Biomechanics

Bioelectrics

Blood flow through vessels

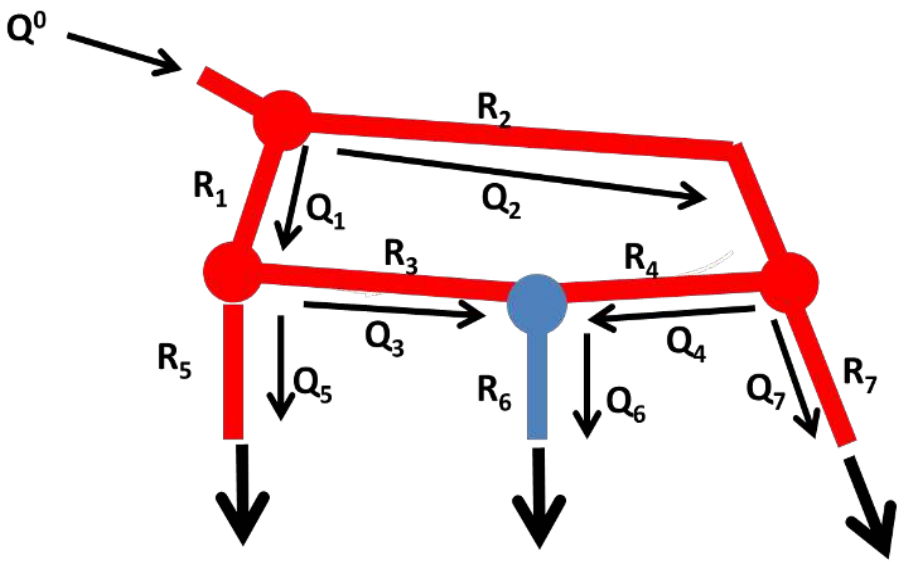
Mass transport

PS #5: Modeling redistribution of blood flow after loss of an artery



Data from lab of Rich Price,
Heuslein et al ATVB 2015

PS #5: Modeling redistribution of blood flow after loss of an artery



$$Q_1 + Q_2 = Q^0$$

$$R_1Q_1 - R_2Q_2 + R_3Q_3 - R_4Q_4 = 0$$

$$R_3Q_3 - R_5Q_5 + R_6Q_6 = 0$$

$$-R_4Q_4 - R_6Q_6 + R_7Q_7 = 0$$

$$Q_5 + Q_6 + Q_7 = Q^0$$

$$Q_3 + Q_4 = Q_6$$

$$Q_4 + Q_7 = Q_2$$

Data from lab of Rich Price,
Heuslein et al ATVB 2015

R2 goes from 2 to 100

Graphical Strategies

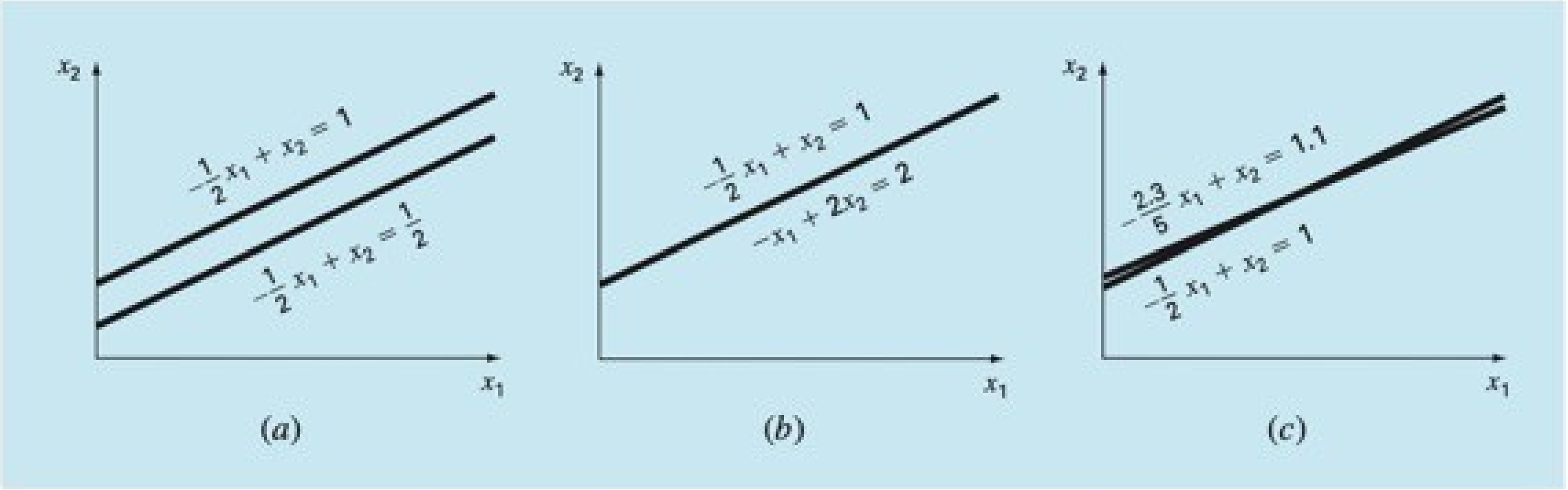
- Solutions to systems of equations are where they intersect

Eq. 1 $3x_1 + 2x_2 = 8$

Eq. 2 $1.5x_1 - x_2 = -1$

More complicated system would be difficult to solve using this method

Graphical strategies



No solutions

Infinite solutions

Ill-conditioned

Singular

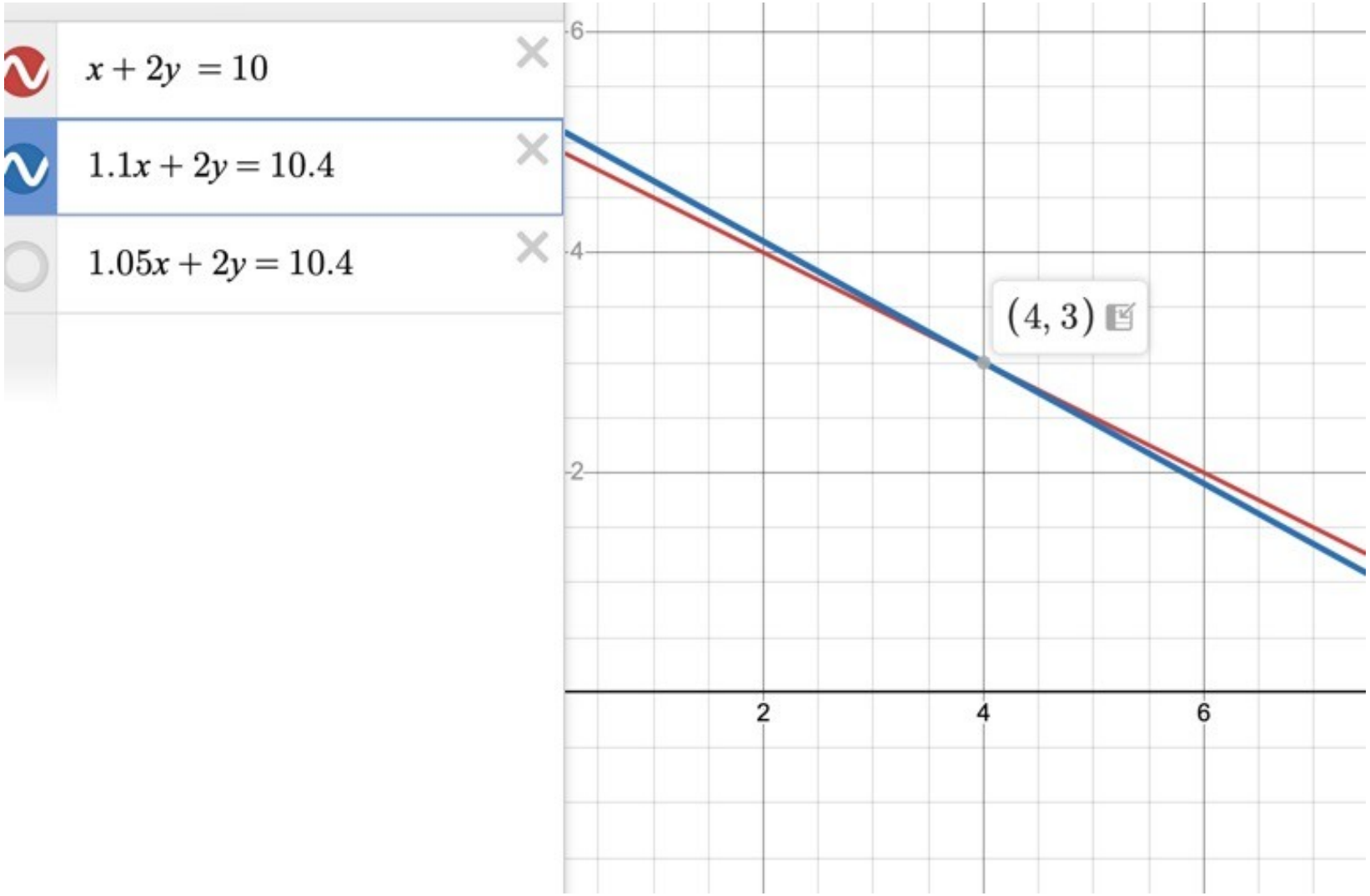
Determinant = 0

Determinant = very small

Ill-conditioned systems

https://www.desmos.com/calculator/jcg5g9ionl

Changing the input a little bit changes the output a lot!



Algebraic Strategies

- Can solve an equation for one variable and plug into another equation

Eq. 1 $3x_1 + 2x_2 = 8$

Eq. 2 $-1.5x_1 + x_2 = 1$

- Can add two equations to eliminate variables

Eq. 1 $3x_1 + 2x_2 = 8$

Eq. 2 $-3x_1 + 2x_2 = 2$

- Can multiply equations by constants

Eq. 1 $3x_1 + 2x_2 = 8$

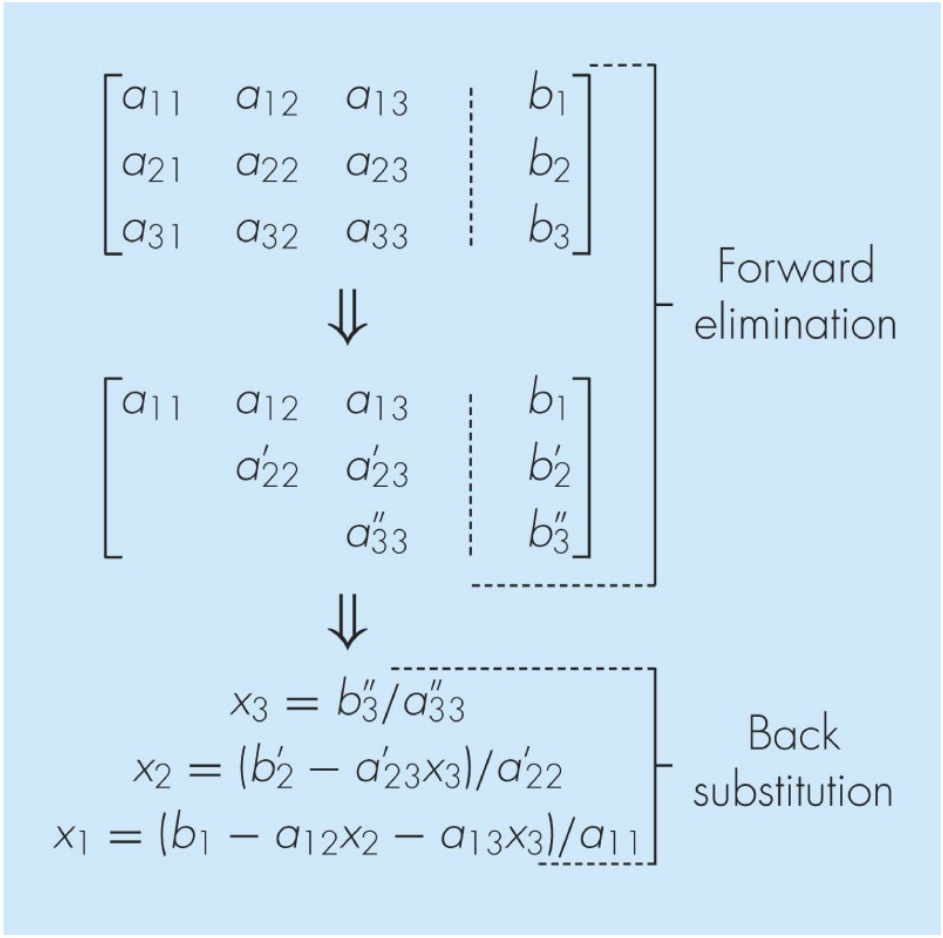
Eq. 2 $1.5x_1 - x_2 = -1$

Gaussian Elimination Overview

Gaussian Elimination generalizes this approach to solve for [x] for large [A] [x]=[b].

1) Augment matrices [A|b]
2) n-1 rounds of forward elimination
3) Back substitution

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Gaussian Elimination: Augmented Matrix

- For Ax=b, make the augmented matrix [A|b]

$$
\begin{aligned}
row_1&: a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\
&\quad a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2 \\
&\quad \vdots \qquad \vdots \qquad \vdots \qquad\qquad \vdots \qquad \vdots \\
row_n&: a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n = b_n
\end{aligned}
$$

$$
\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}
\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} =
\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}
\longrightarrow
\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & & a_{2n} & b_2 \\ \vdots & & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right]
$$

Gaussian elimination steps will be performed on matrix rows

Gaussian Elimination: Forward Elimination, Round 1

$$
\begin{aligned}
row_1&: a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\
row_2&: a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2 \\
row_3&: a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \cdots + a_{3n}x_n = b_3 \\
&\vdots \qquad\quad \vdots \qquad\quad \vdots \qquad\quad\quad \vdots \qquad \vdots \\
row_n&: a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \cdots + a_{nn}x_n = b_n
\end{aligned}
$$

Goal: Eliminate x_1 from equations 2 to n:

$$row_2' = row_2 - \frac{a_{21}}{a_{11}} \cdot row_1$$

$$row_3' = row_3 - \frac{a_{31}}{a_{11}} \cdot row_1$$

$$row_n' = row_n - \frac{a_{n1}}{a_{11}} \cdot row_1$$

In general, for round 1 forward elimination: $row_i' = row_i - \frac{a_{i1}}{a_{11}} \cdot row_1$

Example 1

$$\begin{bmatrix} 2 & -6 \\ -1 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -18 \\ 40 \end{bmatrix}$$

Perform forward elimination, round 1: $row_i{}' = row_i - \frac{a_{i1}}{a_{11}} \cdot row_1$

Gaussian Elimination: Forward Elimination, Round 2

$$
\begin{aligned}
row_1&: \; a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\
row'_2&: \quad\quad a_{22}'x_2 + a_{23}'x_3 + \cdots + a_{2n}'x_n = b_2' \\
row'_3&: \quad\quad a_{32}'x_2 + a_{33}'x_3 + \cdots + a_{3n}'x_n = b_3' \\
&\quad\quad\quad \vdots \quad\quad\quad \vdots \quad\quad\quad\quad \vdots \quad\quad \vdots \\
row'_n&: \quad\quad a_{n2}'x_2 + a_{n3}'x_3 + \cdots + a_{nn}'x_n = b_n'
\end{aligned}
$$

Goal: Eliminate x_2 from equations 3 to n:

$$row_3'' = row_3' - \frac{a_{32}'}{a_{22}'} \cdot row_2'$$

$$row_n'' = row_n' - \frac{a_{n2}'}{a_{22}'} \cdot row_2'$$

In general, for round 2 forward elimination:

$$row_i'' = row_i' - \frac{a_{i2}'}{a_{22}'} \cdot row_2'$$

Gaussian Elimination: Forward Elimination, Cont'd

$$
\begin{aligned}
row_1&: \quad a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\
row_2&: \quad a_{22}'x_2 + a_{23}'x_3 + \cdots + a_{2n}'x_n = b_2' \\
row_3&: \quad a_{33}''x_3 + \cdots + a_{3n}''x_n = b_3'' \\
&\qquad\qquad \vdots \qquad\qquad \vdots \\
row_n&: \quad a_{nn}^{(n-1)}x_n = b_n^{(n-1)}
\end{aligned}
$$

Continue shifting pivot rows until all variables are eliminated up to x_n in the nth equation

In general, for round *k* of forward elimination:

$$row_i^{(k)} = row_i^{(k-1)} - \frac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}} \cdot row_k^{(k-1)}$$

Determinant of triangular matrix = product of diagonal elements

Example 2

$$\begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \\ 6 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}$$

Perform forward elimination.

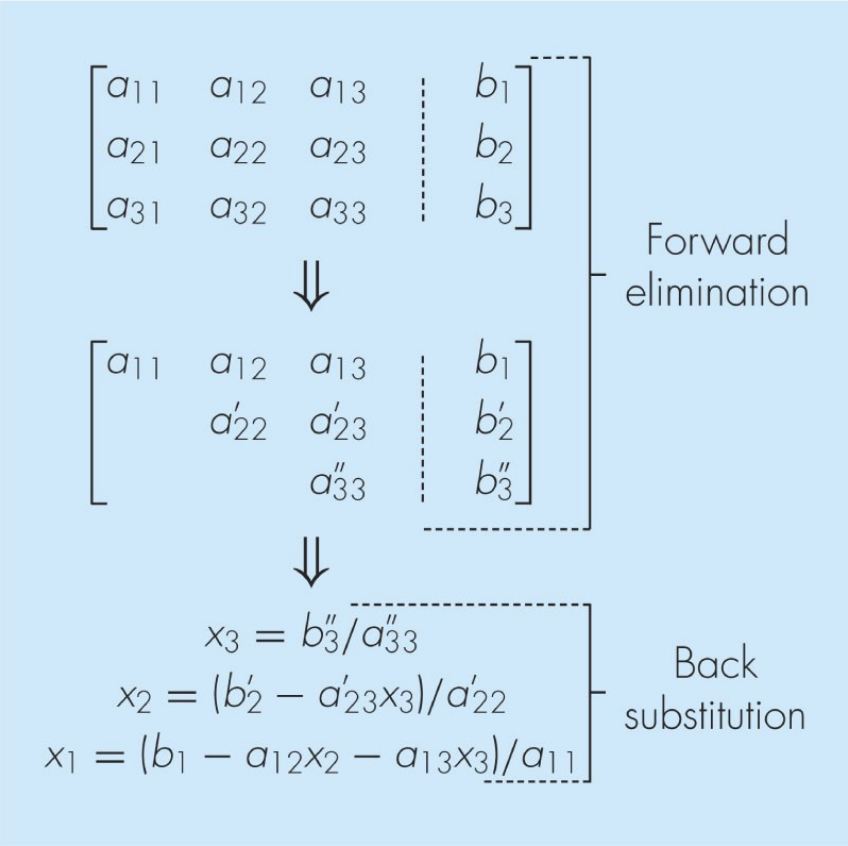
Round 1: $row_i{}' = row_i - \frac{a_{i1}}{a_{11}} \cdot row_1$

Round 2: $row_i{}'' = row_i{}' - \frac{a_{i2}{}'}{a_{22}{}'} \cdot row_2{}'$

Gaussian Elimination: Back Substitution

For n = 3:

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In general:

$$x_n = \frac{b_n^{(n-1)}}{a_{nn}^{(n-1)}}$$

$$x_i = \frac{b_i^{(i-1)} - \sum_{j=i+1}^{n} a_{ij}^{(i-1)} x_j}{a_{ii}^{(i-1)}}$$

for i = (n-1), (n-2), …, 1

Example 1

$$\begin{bmatrix} 2 & -6 \\ -1 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -18 \\ 40 \end{bmatrix}$$

Compute the back substitution to solve for x.

After forward elimination: $\left[\begin{array}{cc|c} 2 & -6 & -18 \\ 0 & 5 & 31 \end{array}\right]$

$$x_n = \frac{{b_n}^{(n-1)}}{{a_{nn}}^{(n-1)}}$$

$$x_i = \frac{{b_i}^{(i-1)} - \sum_{j=i+1}^{n} {a_{ij}}^{(i-1)} x_j}{{a_{ii}}^{(i-1)}}$$

Example 2

$$\begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

Compute the back substitution to solve for x.

After forward elimination: $\left[\begin{array}{ccc|c} 4 & 1 & -1 & -2 \\ 0 & -0.25 & 3.25 & 6.5 \\ 0 & 0 & -4 & -4 \end{array}\right]$

$$x_n = \frac{{b_n}^{(n-1)}}{{a_{nn}}^{(n-1)}}$$

$$x_i = \frac{{b_i}^{(i-1)} - \sum_{j=i+1}^{n} {a_{ij}}^{(i-1)} x_j}{{a_{ii}}^{(i-1)}}$$

Gaussian Elimination Pseudocode

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(a)

```
DOFOR k = 1, n − 1
 DOFOR i = k + 1, n
  factor = a_{i,k} / a_{k,k}
  DOFOR j = k + 1 to n
   a_{i,j} = a_{i,j} − factor · a_{k,j}
  END DO
  b_i = b_i − factor · b_k
 END DO
END DO
```

(b)

```
x_n = b_n / a_{n,n}
DOFOR i = n − 1, 1, −1
  sum = b_i
  DOFOR j = i + 1, n
   sum = sum − a_{i,j} · x_j
  END DO
  x_i = sum / a_{i,i}
END DO
```

Caveats to (Naïve) Gauss Elimination

- Division by zero
  - It is possible to divide by 0 if any pivot coefficients are 0
  - → pivoting: use the largest available coefficient in the column as the pivot point
- Round-off errors
  - Important especially when you have large numbers of equations
  - → Use more sig figs (and use pivoting)
- Ill-conditioned systems
  - Determinant is close to zero (or equal to zero)-
  - → Scaling 1 or more equations increases the determinant
  - → Use consistent units across equations so coefficients are on the same order of magnitude
  - If det = 0, Gauss elimination should be stopped

Determinant of triangular matrix = product of diagonal elements

Key Points

- Mathematical models of biological systems can be represented as linear systems
- Linear systems can be expressed in matrix form as [A][x]=[b]
- Gaussian Elimination
  - Solves [x]=inv([A])*[b]
  - Eliminate variables through forward elimination
  - Back substitution to solve for variables